Group Topic: **Big Data Analysis for Rainfall-Based Flood Risk Prediction in Albay using SYNOP Data**	Lab No. 2
Git Repo/ Colab Link: https://github.com/PotatoKevz/About-bda	Date: 04/24/26

Objectives

- Conceptualize the data gathering method
- Look for online sites that can be requested for SYNOP data
- Decode the downloaded raw data reports in a CSV format
- Clean the decoded data reports
- Visualize the cleaned data reports

Tasks Completed

1. **Looked for sites that can be requested by SYNOP Dataset, and the group arrived at OGIMET (*Organización Meteorológica Iberoamericana)*, an online platform that provides access to historical weather observations, especially SYNOP reports from meteorological stations worldwide.**

2. **Scraped data from weather stations near Albay; Albay (Lezapi City), Camarines Norte (Daet), Camarines Sur (Naga), Catanduanes (Virac), Masbate, Romblon, and Quezon (Infanta and Tayabas) Stations:**

98427 Tayabas (Quezon) 75k.csv	jab2023-6181-30913	Apr 23	8.2 MB	⋮
98434 Infanta (Quezon) 75k.csv	jab2023-6181-30913	Apr 23	8.2 MB	⋮
98440 Camarines Norte 76k.csv	jab2023-6181-30913	Apr 23	8.5 MB	⋮
98442 Camarines Sur Station 11k.csv	jab2023-6181-30913	Apr 23	3.5 MB	⋮
98444 Albay Station 76k.csv	jab2023-6181-30913	Apr 23	8.4 MB	⋮
98446 Catanduanes 76k.csv	jab2023-6181-30913	Apr 23	8.2 MB	⋮
98536 Romblon 74k.csv	jab2023-6181-30913	Apr 23	7.9 MB	⋮
98543 Masbate 76k.csv	jab2023-6181-30913	Apr 23	8 MB	⋮

Scraped Datasets from OGIMET

	A	B	C	D	E	F	G
1	WMOIND	YEAR	MONTH	DAY	HOUR	MIN	REPORT
2	98444	2000	1	1	0	0	AAXX 01001 98444
3	98444	2000	1	1	3	0	AAXX 01031 98444
4	98444	2000	1	1	6	0	AAXX 01061 98444
5	98444	2000	1	1	9	0	AAXX 01091 98444
6	98444	2000	1	1	12	0	AAXX 01121 98444
7	98444	2000	1	1	15	0	AAXX 01151 98444
8	98444	2000	1	1	18	0	AAXX 01181 98444
9	98444	2000	1	2	0	0	AAXX 02001 98444
10	98444	2000	1	2	3	0	AAXX 02031 98444
11	98444	2000	1	2	6	0	AAXX 02061 98444
12	98444	2000	1	2	9	0	AAXX 02091 98444
13	98444	2000	1	2	12	0	AAXX 02121 98444
14	98444	2000	1	2	15	0	AAXX 02151 98444
15	98444	2000	1	2	18	0	AAXX 02181 98444
16	98444	2000	1	2	21	0	AAXX 02211 98444
17	98444	2000	1	3	0	0	AAXX 03001 98444
18	98444	2000	1	3	3	0	AAXX 03031 98444
19	98444	2000	1	3	6	0	AAXX 03061 98444
20	98444	2000	1	3	9	0	AAXX 03091 98444
21	98444	2000	1	3	12	0	AAXX 03121 98444
22	98444	2000	1	4	0	0	AAXX 04001 98444
23	98444	2000	1	4	3	0	AAXX 04031 98444
24	98444	2000	1	4	6	0	AAXX 04061 98444
25	98444	2000	1	4	9	0	AAXX 04091 98444
26	98444	2000	1	4	12	0	AAXX 04121 98444
27	98444	2000	1	4	15	0	AAXX 04151 98444

CSV Content of Albay Station (Raw SYNOT Data Report)

C: > Users > romeb > Desktop > Big Data Analysis > Programs > flood-prediction > ogimet_synop_scraper.py > _chunks

```python
BASE_URL = "http://www.ogimet.com/cgi-bin/getsynop"
COLUMNS  = ["WMOIND", "YEAR", "MONTH", "DAY", "HOUR", "MIN", "REPORT"]

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(message)s",
    datefmt="%Y-%m-%d %H:%M:%S",
)
log = logging.getLogger(__name__)

# Mimic a real browser — OGIMET rejects bot-like user agents
HTTP_HEADERS = {
    "User-Agent": (
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
        "AppleWebKit/537.36 (KHTML, like Gecko) "
        "Chrome/124.0.0.0 Safari/537.36"
    ),
    "Accept"          : "text/plain, text/html, */*",
    "Accept-Language" : "en-US,en;q=0.9",
    "Connection"      : "keep-alive",
}


# — helpers ————————————————————————————————————————————————————————————

def _fmt(dt: datetime) -> str:
    return dt.strftime("%Y%m%d%H%M")


def _chunks(start_year: int, end_year: int, days: int) -> Iterator[Tuple[datetime, datetime]]:
    """Yield (begin, end) datetime pairs in `days`-sized steps."""
    now   = datetime.now(timezone.utc).replace(tzinfo=None)
```

Code Snippet of Data Scraper

3. Cleaned and decoded the SYNOP datasets:

CleanedAlbay.csv	jab2023-6181-30913	6:37 PM	5.4 MB	⋮
CleanedCamNorte.csv	jab2023-6181-30913	6:42 PM	5.4 MB	⋮
CleanedCatanduanes.csv	jab2023-6181-30913	6:42 PM	5.3 MB	⋮
CleanedInfanta.csv	jab2023-6181-30913	6:43 PM	5.3 MB	⋮
CleanedMasbate.csv	jab2023-6181-30913	6:43 PM	5.4 MB	⋮
CleanedRomblon.csv	jab2023-6181-30913	6:43 PM	5.2 MB	⋮
CleanedTayabas.csv	jab2023-6181-30913	6:44 PM	5.3 MB	⋮

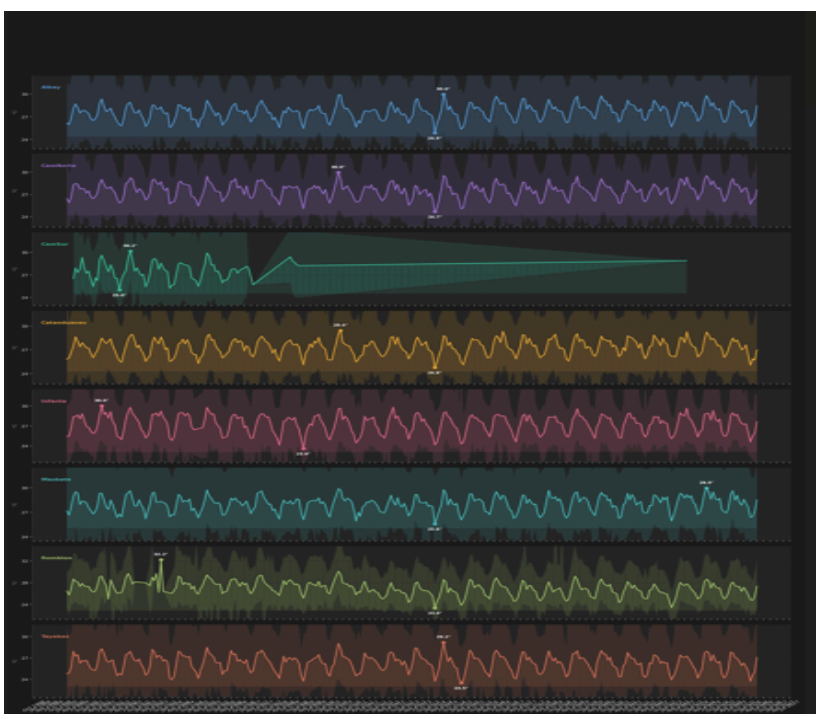
Cleaned Datasets

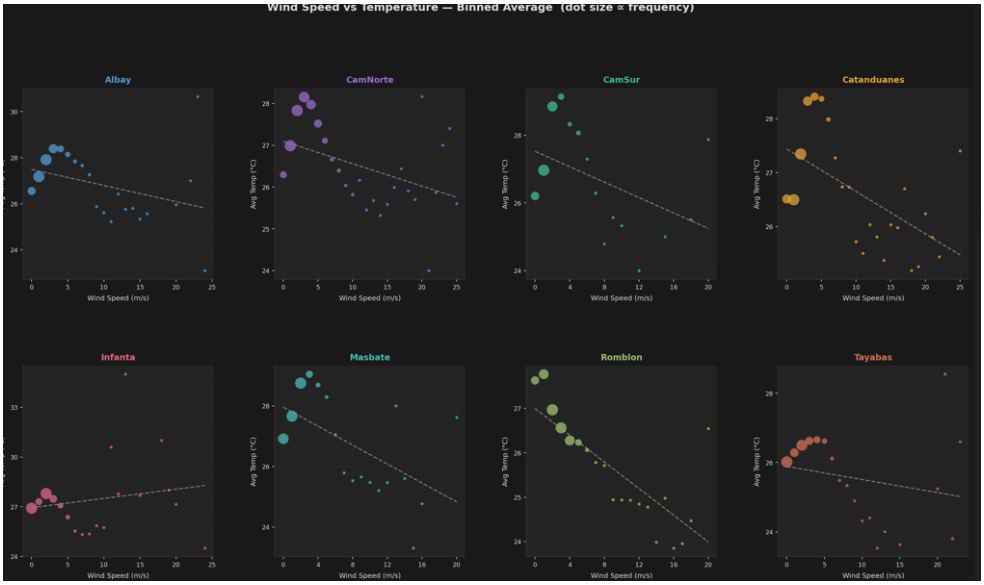
C: > Users > romeb > Desktop > Big Data Analysis > Programs > flood-prediction > synop_decoder.py > ...

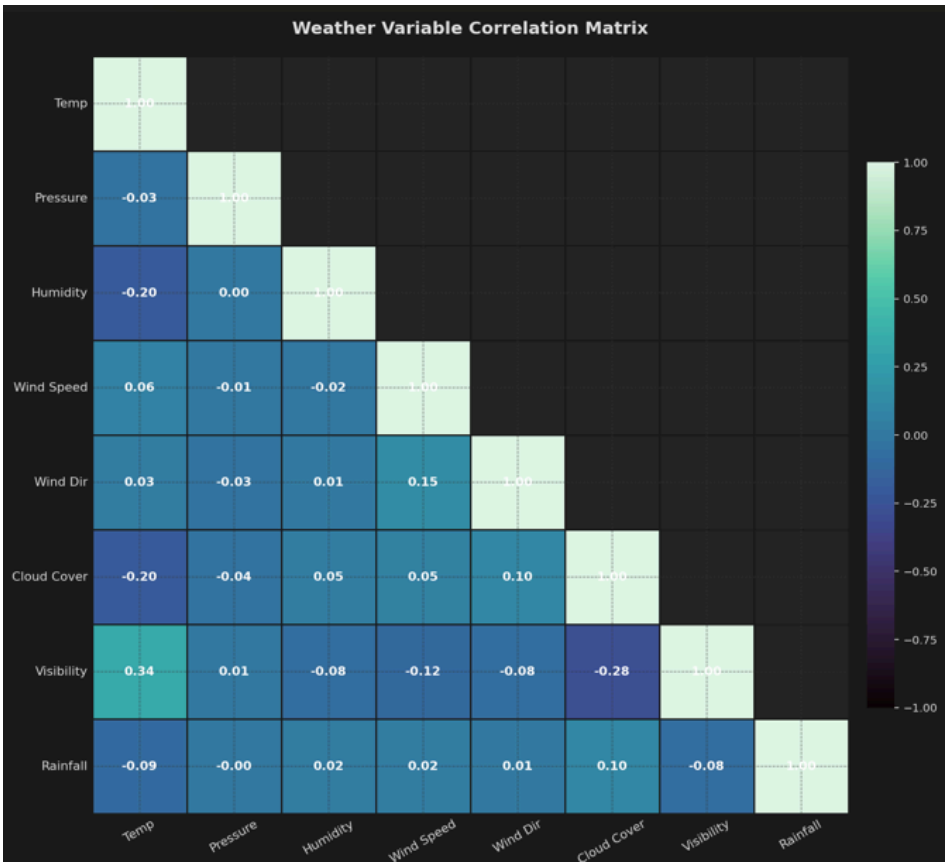
```python
# ----------------------------------------------------------------------------


def decode_irix(group: str) -> dict:
    """Section 0 indicator group  iRiXhVV  (5 chars).

    iR (precipitation indicator):
        0/1/2 - group 6 present somewhere  → no sentinel (use decoded value)
        3     - precip amount = 0, no group 6  → rain_sentinel = 0.0  (Case B)
        4     - station does not measure precip → rain_sentinel = None (Case A)
    """
    out = {}
    if len(group) != 5:
        return out
    ir = group[0]
    if ir == "3":
        out["rain_sentinel"] = 0.0   # "no rain reported" → 0.0 mm
    elif ir == "4":
        out["rain_sentinel"] = None  # truly missing (station doesn't measure)
    # iX - station type / present-weather indicator  (ignored)
    # h  - height of cloud base (code table 1600)
    h_code = group[2]
    h_table = {"0": 0, "1": 50, "2": 100, "3": 200, "4": 300,
               "5": 600, "6": 1000, "7": 1500, "8": 2000, "9": 2500, "/": None}
    out["cloud_base_m"] = h_table.get(h_code)
    # VV - visibility (code table 4377)
    vv = group[3:5]
    out["visibility_m"] = _decode_vv(vv)
    return out


def _decode_vv(vv: str) -> float | None:
    """Convert VV code (00-99) to metres (WMO Code Table 4377)."""
```

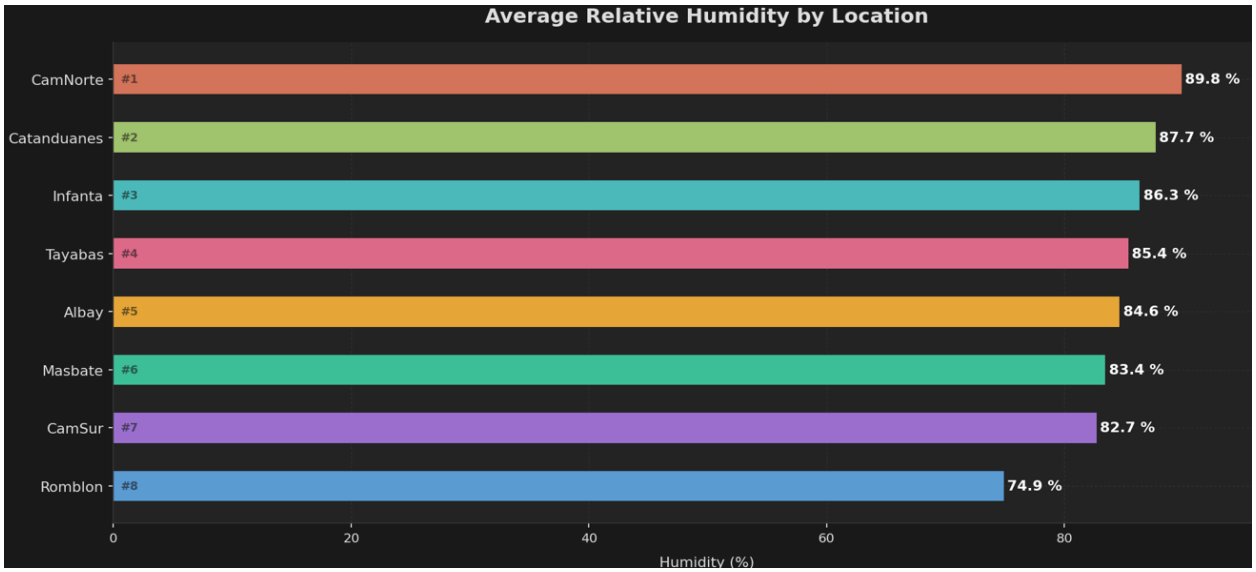
Code Snippet

4. Visualization of the datasets:




Wind Speed vs Temperature — Binned Average (dot size ∝ frequency)
Albay
CamNorte
CamSur
Catanduanes
Infanta
Masbate
Romblon
Tayabas
Avg Temp (°C)
Wind Speed (m/s)


Weather Variable Correlation Matrix
Temp
Pressure
Humidity
Wind Speed
Wind Dir
Cloud Cover
Visibility
Rainfall
1.00
-0.03
-0.20 0.00
0.06 -0.01 -0.02
0.03 -0.03 0.01 0.15
-0.20 -0.04 0.05 0.05 0.10
0.34 0.01 -0.08 -0.12 -0.08 -0.28
-0.09 -0.00 0.02 0.02 0.01 0.10 -0.08
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00



Results and Analysis

Summary of Findings

- **Thermal Stability:** All eight stations maintain a consistent tropical range between 26.3°C (**TAYABAS**) and 28.0°C (**MASBATE**), with a clear seasonal peak in May - June (~28.7°C).

- **Regional Anomalies:** January 2014 was identified as a significant regional cold event, showing record lows across five different stations simultaneously.

- **Humidity & Geography:** Humidity levels are high across the mainland (82% - 90%) but **ROMBLON** acts as a significant outlier **74.9%,** likely due to its island geography and rain-shadow effects.

- **Wind Patterns:** Mean daily wind speeds peak during the Northeast Monsoon (Amihan) in December and January. The data accurately reflected extreme weather, such as the **60.0m/s** gust recorded in Camarines Sur in September 2000.

Challenges and Solutions

Challenge: To look for an efficient, free, and readily available source of SYNOP Data.

Solution: Brainstormed for sources such as NASA, PAGASA, but finally arrived at OGIMET, due to its availability, and tabled report of SYNOP Data.

Challenge: Figuring out the most relevant stations for data gathering, as well as reaching the minimum 500k rows of datasets.

Solution: Researched and mapped the most plausible proximity weather stations of Albay, and arrived to the listed stations. And more stations meant more data to be gathered from.

Challenge: Raw reports are in some OGIMET Format that is very hard to interpret and understand, and some rows are missing values.

Solution: Led the way for a decoder for the raw data that also cleans the CSV at the same time.

Conclusion

The overall tasks successfully moved the project from the planning stage to active data management. By brainstorming data sources, picking relevant data stations, as well as establishing an automated pipeline to scrape and decode SYNOP data, the group has secured the necessary high-fidelity information required for rainfall-based flood prediction. The objectives for data conceptualizing, data gathering, decoding, and initial visualization were fully met.